Bittle X Pi Add-on v0.16 - Build Guide

Version 0.16, 2026-10-15. A sample add-on for the Bittle X platform v3.1 (../bittle_platform). It holds a Raspberry Pi Zero 2 W with a PiSugar 2 (1200 mAh), an NS4168 I2S amp with a 3 W speaker, and an SPH0645 I2S mic. The case is one rounded box (the body) and a snap-in top panel. The body clicks onto the platform pegs, and every part goes in from the top. No tools, glue, tape or supports are necessary.



The add-on on the platform: the mic holes are at the front, the port windows in the panel.

1. Parts to print

File (stl/print/)	Material	Orientation	Supports	Cura profile	Mass
pod_body.stl	PLA	floor on the bed	none	Bittle Pi Pod - PLA 0.25	32 g
pod_panel.stl	PLA, second color	upside down: top on the bed	none	Bittle Fine 0.10 for the best top	13 g
pod_switch.stl	PLA	flat side on the bed, tab up	none	Bittle Fine 0.10	0.2 g
cover_pegs_foot.stl (../bittle_platform)	PLA	stand on the rear end	none: the foot is the support	Bittle Cover - PLA 0.25	11 g

Print quality. The body walls are vertical and its floor is on the bed. The only overhangs (30 mm² in total) are the tops of the two cable notches at the base of the side walls: 8 mm bridges. The front cable notches are 45 deg chamfers. The grille holes and the mic holes are teardrops (round, with a 45 deg point on top), so they print without bridges. The panel top edge round ends in a 45 deg chamfer at the bed, the port openings have 45 deg chamfers, and the snap bumps have 45 deg catch faces. Its only overhang (4 mm²) is the 2 mm bridge of the button arm. The slider has a 1.7 mm bridge over the fork. Keep the seam at the rear.

2. Electronics

Part	Place
Raspberry Pi Zero 2 W + PiSugar 2, 1200 mAh	On the long edge over the robot center: ports and PiSugar controls up, GPIO header down to the right, SD card to the front. Two end posts with grooves hold the boards.
Dupont jumpers	On the soldered Pi header, 12 pins (section 4). 3 of them go to the BiBoard header in the right well.
NULLLAB NS4168 I2S amp	38.4 x 22.4 mm board, on edge along the left wall, parts to the inside, in a slot bar on the floor. Its PH2.0 connectors are side entry: the I2S cable plugs in at the front end, the speaker at the rear end.
NULLLAB 3 W 4 ohm speaker	At the rear, sound to the grille in the rear face. Ear plate 45 x 31 x 1.5 in two side grooves (ruler).
HiLetgo SPH0645 I2S mic	16.7 x 12.7 x 1.8 board at the front wall, port to the front, in two grooves. Solder the wires to the pins at its top edge.
Head camera cable	The Grove cable comes out under the front edge of the cover (left side) and runs up to the head. A 45 deg notch in each front bottom corner of the body gives it room.
Optional cable	A notch at the base of each side wall lets a cable in from outside.

Balance. The add-on weighs about 116 g (printed parts in PLA and typical masses of the electronics, not weighed). The Pi stack, the heaviest part, is over the robot center. The speaker is at the rear, away from the mic. So the center of mass of the add-on is 13.5 mm behind the robot center and 2.6 mm left of the center line, 19 mm above the cover top. The platform spec recommends 10 mm or less: with the speaker at the front it is about 7 mm. Put your own masses in MASS_G in pod.py and

run pod.py again.

3. Assembly

The 3D viewer shows each step with animation: https://beetlebugorg.github.io/bittlex/viewer/#step=1.

1 Fit the platform cover in place of the stock cover. (viewer step 2)

2 Put the empty body on the pegs 7 mm behind its place, grille to the rear. Slide it 7 mm forward until it clicks (about 4.2 N). The BiBoard header is now in the notch of the right well. (viewer step 3) (viewer step 4)

3 Push the speaker down into the two side grooves at the rear, body to the rear. (viewer step 5)

4 Push the mic down into the two grooves at the front wall, port to the front. (viewer step 6)

5 Plug the amp cable into the I2S connector at the front end of the amp and the speaker into its rear end. Push the amp down into the slot along the left wall, parts to the inside. Lay the amp cable across the floor in front of the front pegs. (viewer step 7)

6 Push the jumpers onto the Pi header. Push the Pi + PiSugar 2 stack down into the end grooves, over the I2S wires. (viewer step 8)

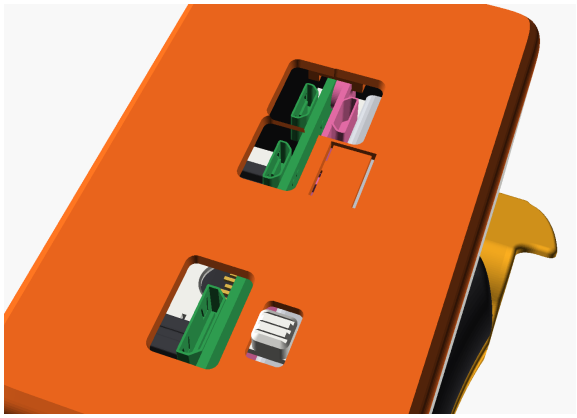
7 Reach into the right well and push the three Dupont plugs onto the BiBoard header (TX2, RX2, GND). Fold the slack into the right well. (viewer step 9)

8 Put the switch slider on the PiSugar switch lever, fork down. (viewer step 10)

9 Push the panel straight down until the 4 tabs click (about 9.1 N). It fits one way only. The slider tab comes up through its slot. (viewer step 11)

10 Open: lift the rear edge of the panel at the finger notch (about 8 N), then the front. Remove: open the panel, pull the BiBoard plug off, slide the body 7 mm rearward and lift it.

On the top: the port openings are shaped to the plug bodies (plug + 0.4 mm). The Pi power port and the PiSugar charge port share one opening. The ribbed slider beside the HDMI opening moves the PiSugar power switch (2 mm of slot travel each way; the switch sets the end positions). The tongue near the front of the panel is a large button: press it to press the PiSugar button.



The top: HDMI, the switch slider, data USB, charge and Pi power, and the button tongue.

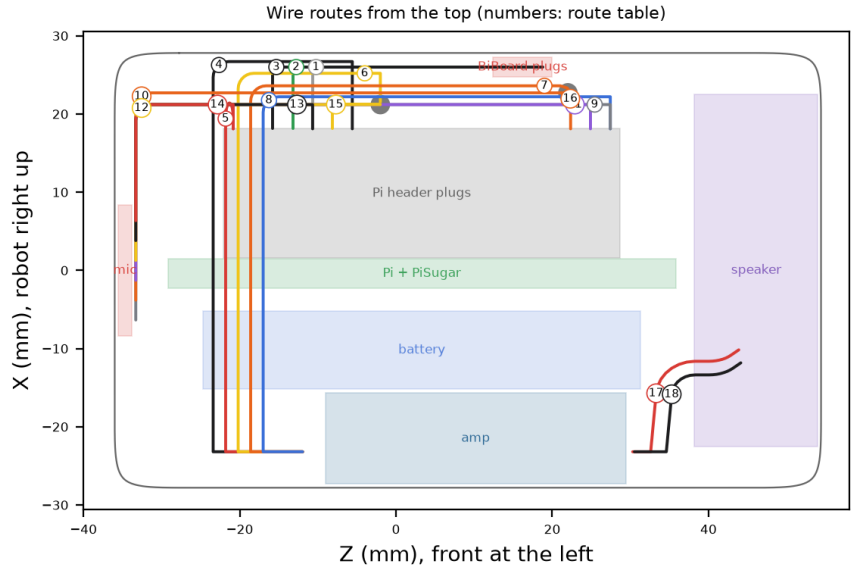
4. Wiring

Board pin	Pi pin	Signal
BiBoard TX2 / RX2 / GND	10 / 8 / 6	RXD / TXD / GND (TX and RX cross). BiBoard +5V: not wired
Amp V / G	2 / 14	5V / GND
Mic 3V / GND / SEL	1 / 9 / 39	3V3 / GND / GND (left channel)
Amp BCLK + mic BCLK	12 (splice)	GPIO18: one wire from the pin to a soldered splice, then one wire to each board
Amp LRC + mic LRCL	35 (splice)	GPIO19: the same, a second splice
Amp DIN / mic DOUT	40 / 38	GPIO21 / GPIO20
Speaker	-	amp 2-pin PH2.0 at the rear end of the amp

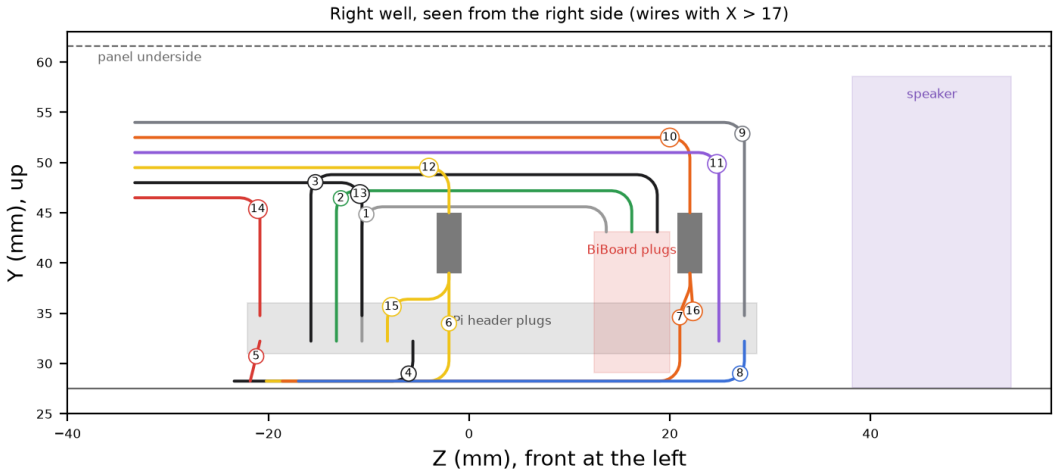
Pin map from the wiring blueprint (reference/g2-wiring-blueprint.html; that blueprint wins if they differ). PiSugar 2 touches pins 2, 4, 5 and 6 from below with its pogo pins. Pins 11, 17-20 and 22-24 are kept free for a screen.

5. Cable routing

pod.py models every wire (1.4 mm) on its route and checks it: no wire touches a printed part or a board, no two wires pass through each other (0 found), and no bend is smaller than 2 mm radius. The drawings show the routes. The numbers are the rows of the table.



Top view of the routes. The right well is between the Pi header plugs and the right wall.



The right well from the right side: the floor lanes of the amp cable, the BiBoard wires at three heights, the mic cable above them.

#	From	To	mm	#	From	To	mm
1	BiBoard TX2	pin 10	46	10	mic LRCL	LRCLK splice	99
2	BiBoard RX2	pin 8	54	11	mic DOUT	pin 38	110
3	BiBoard GND	pin 6	62	12	mic BCLK	BCLK splice	62
4	amp G	pin 14	104	13	mic GND	pin 9	60
5	amp V	pin 2	72	14	mic 3V	pin 1	45
6	amp BCLK	BCLK splice	96	15	BCLK splice	pin 12	13
7	amp LRC	LRCLK splice	111	16	LRCLK splice	pin 35	8
8	amp DIN	pin 40	107	17	amp speaker +	speaker +	38
9	mic SEL	pin 39	121	18	amp speaker -	speaker -	38

1 Mic wires: bend each wire before you solder it, so it leaves along the back face of the board. Cut them to the route length + 15 mm. Route: up the back of the mic, right over the front post, rearward down the right well; the lowest wire leaves first.

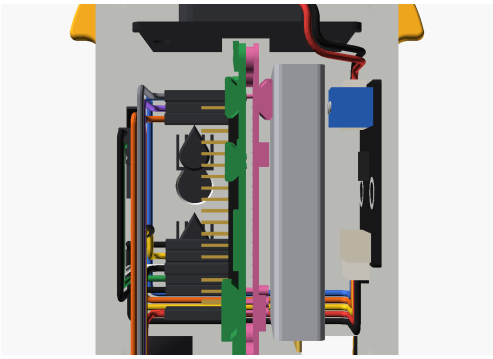
2 Amp cable: forward from the plug, down, across the floor in front of the front pegs (under the stack), then rearward in a floor lane of the right well. From the front: DIN in the inner lane, then LRC, BCLK, and G at the wall. V goes up to pin 2 at once. Lay it before the stack goes in.

3 Splices: the BCLK splice stands in front of the BiBoard notch, the LRCLK splice beside the BiBoard plugs. The Pi wire and the amp wire go into the bottom, the mic wire into the top.

4 BiBoard wires: right well: along the floor to the right wall, up, rearward over the BiBoard plugs (each wire at its own height), down into them. Pin 6 (front) goes to GND (rear), so the order turns over: pin 10 runs lowest, pin 6 highest.

5 Speaker leads: a pair: up the left side behind the amp, over the ear plate, into the top of the speaker.

6 Jumpers are longer than the routes. Fold the slack in the right well above the floor lanes, away from the finger space over the BiBoard plugs.



From the top, panel off: the wires in 3D.

6. Checks

1 Slide: the empty body is clear of the robot at the click point, half way and at the drop-on place. The clicks are about 4.2 N to lock and 4.2 N to unlock (beam strain 1.12 %).

2 Panel tabs: 9.1 N to close (4 tabs, 30 deg lead), 8 N to open one end (2 tabs, 45 deg catch), tab strain 1.19 %. Turned 180 deg, the panel does not go down.

3 Install: the speaker, the mic, the amp and the stack each go straight down 40 mm without contact. The panel goes down without contact.

4 Switch and button: the slider moves 2 mm each way without contact, and its fork pushes the lever after 0.3 mm. The button arm is 0.2 mm over the button; pressed 0.6 mm, it pushes the button.

5 Robot: the head turns from -85 to 85 deg without contact. 6 front and 22 rear leg poses (of 504) touch the add-on, only when a thigh swings up over the back. The Petoi tail is 3.2 mm under the add-on, and 0.05 mm at the drop-on place.

Open items: the BiBoard header height is an estimate (measure it). The speaker size is a ruler measurement. The mic port is assumed at the board center; the 7 holes cover ±2.2 mm. The SD card is assumed 2.5 mm out of its holder. To change the SD card, lift the stack out. The Pi power port and the PiSugar charge port are side by side: use one of them at a time.

7. Files and license

File	Content
stl/print/	the parts in print orientation
cad/pod_body.step, cad/pod_panel.step	the parts as STEP (from the mesh)
cad/assembly_step.zip	one STEP file with the robot, the platform and the add-on in place, with named and colored parts
stl/installed/, stl/models/	the parts and the electronics in the robot frame
pod.py, motion.py, render.py, docs.py	dimensions, build and checks; robot motion; pictures; this guide

The printed parts, the drawings and this guide are CC BY 4.0, and the code is MIT (Jeremy Collins). The Pi Zero and PiSugar 2 fit models are from github.com/PiSugar/PiSugar (GPL-3.0). The Petoi parts are from github.com/PetoiCamp/NonCodeFiles, without a stated license. LICENSE.md and ATTRIBUTION.md in the repository give the details.